Eric Xiang Gao

First-year Computer Science PhD Student at the University of Illinois Urbana-Champaign

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Research Interests

RL for Systems and Networking; Agentic AI for Systems; Systems for Agentic AI.

Education

University of Illinois Urbana-Champaign Urbana, IL, USA

PhD student in Computer Science Aug. 2026 – Present

- Advisor: Prof. Francis Y. Yan

Tsinghua University Beijing, China

Undergraduate studies in Computer Science, IIIS (Yao Class) Aug. 2022 – Jun. 2026

- Major GPA: 3.8/4.0.
- Selected coursework: C++ Programming (4.0, A+); Calculus (4.0, A+); Theory of Computation (4.0); Advanced Computer Graphics (4.0); Distributed Systems and Operating Systems (4.0); Game Theory (4.0); Computer Architecture (4.0); Machine Learning (3.6).

Publications

[1] Seoyul Oh, Yikai Zheng, **Eric Xiang Gao**, Deepak Vasisht, and Francis Y. Yan. "Rediscovering Heuristics: A Litmus Test for RL in Networked Systems." *ACM Workshop on Hot Topics in Networks (HotNets)*, 2026.

[2] Minrui Luo, Weihang Xu, **Xiang Gao**, Maryam Fazel, and Simon Shaolei Du. "Global Convergence of Four-Layer Matrix Factorization under Random Gaussian Initialization." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.

[3] Solaleh Mohammadi, **Xiang Gao**, and Kaiqing Zhang. "Entropic Risk-Sensitive Evolutionary Learning and Equilibrium Selection in Coordination Games." *IEEE Conference on Decision and Control (CDC)*, 2026.

[4] Siting Li, **Xiang Gao**, and Simon Du. "Highlighting What Matters: Promptable Embeddings for Attribute-Focused Image Retrieval." *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

[5] Jiaxuan Jiang, **Xiang Gao**, and Zhixuan Fang. "PreACT: Pre-Action Communication of Tentatives for Multi-Agent Coordination and Learning." Preprint, 2025.

Research Experience

Rediscovering Heuristics Urbana, IL, USA

Compact RL policies for classic network-control problems Jun. 2026 – Present

Advisor: Prof. Francis Y. Yan, University of Illinois Urbana-Champaign

- Investigated whether compact RL policies can rediscover the core behavior and performance of classic network heuristics through reward-driven learning alone.
- Developed a diagnostic methodology combining return landscapes, optimization paths, and behavioral correctness checks to explain why RL training succeeds or fails.
- Constructed low-dimensional policy classes for adaptive bitrate streaming (ABR) and congestion control, enabling direct analysis of how training choices affect learned policies.
- Demonstrated that simple policy gradient methods recover buffer-based adaptation rules and approach the performance of tuned heuristic baselines in controlled ABR experiments.

- Identified failure modes caused by unstable optimization, misweighted rewards, and unrepresentative training traces, providing guidance for diagnosing and repairing RL training pipelines.
- Currently exploring extensions to TCP CUBIC and BBR by aligning observations, actions, and decision intervals and isolating control components that compact RL policies can represent.
- Currently studying how to increase policy complexity and training-trace diversity while preserving interpretable behavior and improving generalization to broader network conditions.
- Part of this work is reported in publication [1].

Risk-Sensitive Multi-Agent Reinforcement Learning Remote

Equilibrium selection in coordination games Jul. 2025 – Apr. 2026

Advisor: Prof. Kaiqing Zhang, University of Maryland

- Investigated limitations of traditional equilibrium concepts in coordination games when positive and negative payoffs affect agent behavior asymmetrically.
- Introduced exponential-form risk measures to account for payoff asymmetry and align agent policies with risk-aware objectives.
- Theoretically characterized the sensitivity of Stackelberg equilibrium to changes in the risk parameter in discretized game models.
- Established convergence guarantees to risk-sensitive equilibria under replicator dynamics, softmax policy gradient, and natural policy gradient methods.
- This work is reported in publication [3].

RHO-Loss for VLM2Vec Seattle, WA, USA

Data scheduling for efficient vision-language model training Feb. 2025 – Aug. 2025

Advisor: Prof. Simon Shaolei Du, Paul G. Allen School, University of Washington

- Investigated data scheduling for VLM2Vec to address lengthy training and uneven performance across task categories.
- Applied data scheduling methods, including RHO-Loss, to accelerate pretraining.
- Used CLIP-like teacher models to guide data selection based on the loss difference between the model being trained and a reference model.
- Modified the VLM2Vec codebase to integrate data selection and evaluated the approach using multiple metrics.

Highlighting What Matters Seattle, WA, USA

Promptable embeddings for attribute-focused image retrieval Feb. 2025 – May 2025

Advisor: Prof. Simon Shaolei Du, Paul G. Allen School, University of Washington

- Investigated limitations of CLIP-like models in retrieving images based on selected attributes rather than overall image–text similarity.
- Developed the COCO-Facet benchmarks, on which generative models outperformed contrastive models by more than 10% in text-to-image retrieval.
- Evaluated baseline models and implemented interfaces to adapt them to the benchmark data; observed approximately 5% better retrieval performance with prompting.
- Designed prompts manually and through LLM APIs, and evaluated the improvements achieved by both approaches with generative models.
- Identified retrieval weaknesses in existing models, including limited robustness to deceptive attributes.
- This work is reported in publication [4].

Tentative Anticipatory Intention Communication (TAIC) Beijing, China

Intention-based communication for multi-agent reinforcement learning Oct. 2023 – Feb. 2025

Advisor: Prof. Zhixuan Fang, IIIS, Tsinghua University

- Investigated communication of agents' intended actions to improve cooperation beyond observation-based messaging.
- Trained the message-generation network using discrepancies between observations/actions estimated from messages and their actual values as supervision.
- Evaluated the approach on standard Multi-Agent Particle Environment (MPE) benchmarks, observing improved avoidance of suboptimal policies and performance gains of more than 10% across multiple environments.
- Conducted most of the baseline-model evaluations.
- Developed an additional research interest in the effects of adversarial agents in multi-agent systems.
- This work is reported in publication [5].

Course Projects

Blockchain with Multiple Participants and Functions Beijing, China

Instructor: Prof. Wei Xu, IIIS, Tsinghua University Sep. 2024 – Jan. 2025

- Implemented a blockchain that selects the longest chain as the main chain, with multiple security mechanisms and a graphical user interface.
- Collaborated with Jingyi Lv and Wenjie Huang.

Multicast Implementation in Garnet Beijing, China

Instructor: Prof. Kaisheng Ma, IIIS, Tsinghua University Jun. 2024 – Aug. 2024

- Worked on multicast routing to deliver the same message from one sender to multiple receivers, with the goal of reducing network traffic by replicating packets only at branching points.

Activities and Interests

Academic exchange, outreach, and campus activities Sep. 2022 – Present

- Participated in the 2023 Shanghai Jiao Tong University–Tsinghua University–Peking University Computer Science "Growth Partner" International Summer School, exchanging ideas with scholars and students from China and abroad.
- Participated in the "Exploring Anhui and Visiting Cultural Heritage" summer research camp and contributed to research and publicity activities.
- Participated in international exchanges organized by the campus "Global Competence" organization, including a discussion with Marcos Galvão, then Brazil's ambassador to China.
- Participated in volunteering and choral activities on and off campus; interested in contributing to class and departmental activities.